DSD F14

Homework #4

1. Write a Verilog model for a 74HC192 synchronous 4-bit up/down counter. Don't include any timing data. Make sure you include in the sensitivity list (Down, Up, Clr, Load)

```
module counter_74HC192(
input wire Clk,
input wire Clr,
input wire Load,
input wire Up,
input wire Down,
);
always @(posedge Clk or Down or Up or Clr or Load) begin
if (Clr ==1) begin
Q <= 4'b0000;
end else if (Load) begin
Q <= D;
end else if (Up && !Down) begin
Q <= Q + 1;
end else if (Down && !Up) begin
Q <= Q - 1;
end
end
endmodule
```

2. Write Verilog code for a module describing a 8-bit shift register that shifts bits in from input SI and out an output SO. Also include control signals Clk and EN (and enable active Hi). Thus: You should have 3 inputs SI, Clk, EN and 1 out SO: all these are single bit signals. The shifting causes 1 bit to go in and 1 bit to go out and remembers the other 7 bits, but the bit just shifted in will not get shifted out for 8 more clock edges. Use a rising edge clock.

```
module shift_register_8bit(
input wire Clk,
input wire EN,
input wire SI,
output wire SO
);
reg [7:0] shift_reg = 8'b0;
// Shift on rising edge of clock when enabled
always @(posedge Clk) begin
if (EN) begin
shift_reg <= {shift_reg[6:0], SI};
end
end
// Serial output is the MSB
assign SO = shift_reg[7];
endmodule
```

3. Write a Verilog module for a SR Latch holding 1 bit. Do it 2 ways:

    a. Use an Boolean equation

```
module sr_latch(
input wire S,
input wire R,
output wire Q,
);
assign Q = S | (Q & ~R);
assign Qn = R | (Qn & ~S);
```

Endmodule

b. Use a gate level description

```
module sr_latch_gates(
input wire S,
input wire R,
output wire Q,
output wire Qn
);
nor #1 nor1(Q, R, Qn);
nor #1 nor2(Qn, S, Q);
endmodule
```